Chapter 4

Knowledge Representation

A study material for the students of GLS University

Introduction

➢ Data

- Data refers to **raw, unorganized, and unprocessed facts and figures**.
- Data by itself does not give any meaning. It is just a collection of values or statements.
- Data is **static in nature**, which means it does not change unless processed.
- **Example:** "The price of crude oil is $80 per barrel."
- This statement is only a fact and does not explain any situation.

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Introduction

➢ Information

- Information is **processed data**. When data is analyzed and placed in a proper context, it becomes information. Information helps users to **understand the meaning of data**.
- **Example:**
  "The price of crude oil rises from $80 to $120."
- This statement gives information because it clearly shows that there is a **price increase of $40**, which indicates a rise in crude oil prices.

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Introduction

➢Knowledge

- Knowledge is the **human understanding of a subject** obtained through **study and experience**.
- It helps in analyzing information and taking decisions.
- Knowledge is defined as expertise and skills acquired by a person through experience or education, which provides theoretical or practical understanding of a subject.

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Introduction

➢**Wisdom**

- Wisdom is the **ability to apply knowledge correctly** to make **good and long-term decisions**.
- It goes beyond knowledge and requires **experience, judgment, and continuous learning**.
- Creating wisdom is a **continuous process** of discovering better conclusions and insights.

Example: "If I purchase more shares of Indian Oil company now, it will be beneficial for my future.

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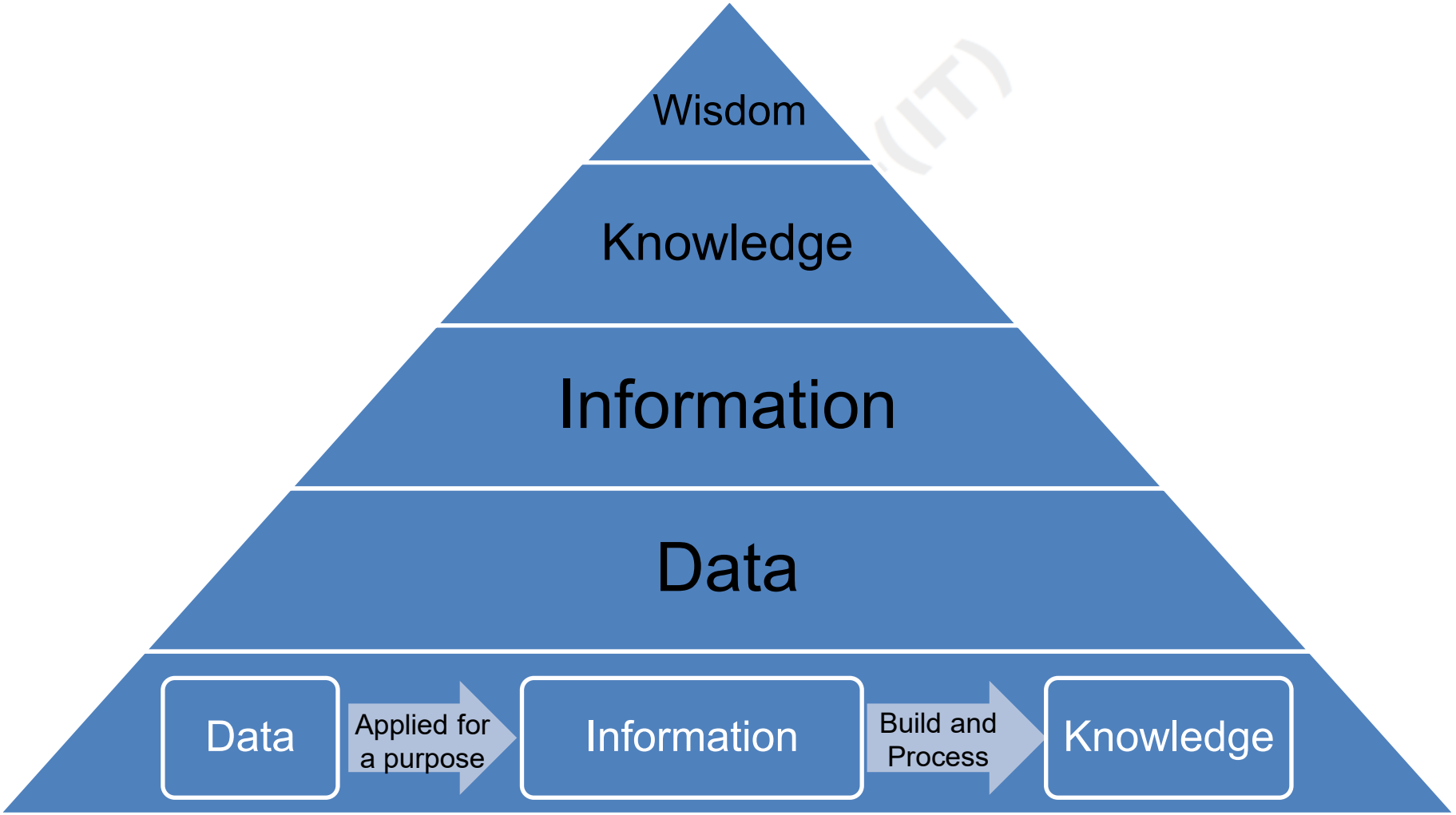
Wisdom
Knowledge
Information
Data
Data
Applied for a purpose
Information
Build and Process
Knowledge

Knowledge Management

- Knowledge Management is a **set of strategies and practices** used in organizations to:
  - ✓Identify knowledge
  - ✓Create knowledge
  - ✓Represent knowledge
  - ✓Distribute knowledge
  - ✓Enable proper use of knowledge and experience
- It helps organizations to **use insights effectively** for decision making.

Important Observations

- Collection of **data** is not information.
- Collection of **information** is not knowledge.
- Collection of **knowledge** is not wisdom.
- Collection of **wisdom** is not truth.
- Each stage requires **processing, understanding, and reasoning**.

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Knowledge Management

Data:

- Raw facts with no meaning
- Example: “The price of Indian Oil is ₹70 per litre.”

Information:

- Processed data with context
- Example: “Today the share value of Indian Oil increased by 1%.”

Knowledge:

- Understanding and prediction from information
- Example: “Crude oil usage is increasing, so Indian Oil share value will increase.”

Wisdom:

- Using knowledge to make future decisions
- Example: “Buying more Indian Oil shares now will benefit me in the future.”

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Value of Knowledge Management

- Suppose a company ABC has a share value of **₹500**.
- If the market value increases by **50%**, we can easily calculate the current share value using data.
- But if we are asked whether the share value will **increase in the next quarter**, we cannot answer using only data and information.
- This happens because we **lack proper knowledge** to predict future trends.
- To make predictions, we need knowledge about:
  - Company's current products
  - Market demand
  - Market trends
  - Competitors
- Knowledge management uses **past and present data** to **predict future share values**.
- Knowledge can be collected through the **Internet** using tools like **crawlers and spiders**, which gather information from different sources.

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Categories of Knowledge

1. Symbolic, Sub-Symbolic, and Hybrid Knowledge

Symbolic Knowledge

- Represented using **symbols and formal languages**
- Stored as **clear and discrete units**
- Easy to understand and explain

Sub-Symbolic Knowledge

- Cannot be represented using single symbols
- Stored in the form of **patterns or information**
- Meaning is obtained through processing

Hybrid Knowledge

- Combines both **symbolic and sub-symbolic** knowledge
- Uses advantages of both representations

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Categories of Knowledge

2. Declarative and Procedural Knowledge

Declarative Knowledge

- Represents **facts** about objects or events
- Describes **what is true**
- Examples:
  - "I am a human."
  - "The sun is shining."

Procedural Knowledge

- Represents **rules, skills, or procedures**
- Describes **how to do something**
- Examples:
  - "If John is a human, then John has two legs."

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Types of Knowledge

Declarative Knowledge

- Declarative knowledge represents **facts about objects and events**.

It is divided into:

- Simple Relational Knowledge
- Inheritable Knowledge
- Inferential Knowledge

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Simple Relational Knowledge

In this type, declarative facts are represented as **relations**.

- Databases are used to support relational knowledge.
- Facts are stored in a **simple format**.
- Each fact is arranged clearly in **rows and columns**.
- There is very little scope for drawing conclusions automatically.
- It mainly works as a **knowledge base for inference engines**.
- The easiest way to represent declarative facts is through **tables**.

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Simple Relational Knowledge

- Example: Information about students in a department can be stored in a table.
- **Major Problem:**
- This method does not represent complex knowledge.
- We need better mechanisms to improve representation.

Name	Class	Marks
Vinod	S3 CS	76
Ramesh	S5 CS	88
Hari	S5 EC	51

Inheritable Knowledge

- Objects inherit properties from their **class**.
- Data is organized in a **hierarchical structure** (tree or graph).
- It uses inheritance property.
- The goal is to represent knowledge using:
  - ✓Specialization (more specific)
  - ✓Generalization (more general)

Structure:

- Nodes represent classes.
- Edges represent relationships.
- Relationships:
  - ✓IS-A (class to class)
  - ✓Instance-of (class to object)

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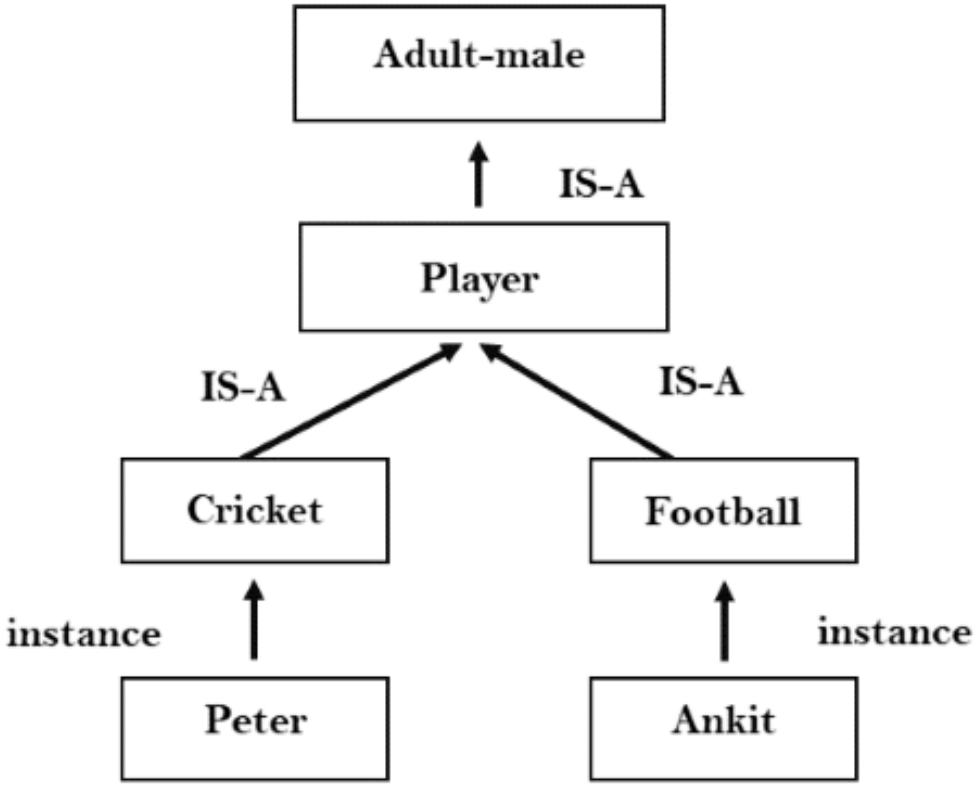
Inheritable Knowledge

- Example:
  A cricket player is a player.
  A player is an adult male.
  So properties are inherited automatically.



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Inferential Knowledge

- Inheritance is a strong form of inference, but not the only one.
- Knowledge is described using **logic**.
- Logical connectives and inference rules are used.
- New knowledge can be derived from existing facts.
- This type of knowledge is used in reasoning systems.

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Procedural Knowledge

- Procedural knowledge represents **how to perform a task**.
- It is represented in the form of programs.
- The machine uses this knowledge while executing code.
- It is defined as algorithms that help infer new knowledge from facts.
- This type of knowledge is computationally efficient,
  but it is difficult to modify and extract new knowledge from it.
- Example:
  DOB = DD-MM-YYYY
  Function Age(DOB)

Procedural Knowledge

Advantages of Procedural Knowledge:

- It is domain specific.
- It supports extended logical inference.

Problems of Procedural Knowledge:

- **Completeness:** Cannot represent all cases.
- **Consistency:** May give incorrect deductions.
  - ✓ Example:
    We know "Tweety is a bird."
    We may conclude "Tweety can fly."
    But later we know "Tweety is an emu."
    Emus cannot fly.
- Modularity is reduced.
- Control information becomes complex.

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Knowledge Representation

AI systems use different ways to represent knowledge.

Types of knowledge used in AI:

- Objects – facts about real world entities.
- Events – actions that occur in the world.
- Performance – behavior and knowledge of how to do something.
- Meta-knowledge – knowledge about knowledge.
- Facts – truths that need to be represented about the real world.

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Approaches to Knowledge Representation

- A good knowledge representation system must satisfy four properties:

1. Representational Adequacy

- Ability to represent all required knowledge in a domain.
- Facts must be properly stored in data structures.
- If representation is weak, it cannot capture detailed meaning.
- Example:
  "John believes no one likes Brussels sprout."
  This is difficult to represent in a simple string format.

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Approaches to Knowledge Representation

2. Inferential Adequacy

- Ability to draw conclusions from represented knowledge.
- Example:
  "If it rains, John never goes out."
  "It is raining today."
  Conclusion: John will not go out.

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Approaches to Knowledge Representation

3. Inferential Efficiency

- Ability to organize knowledge so that inference can be done quickly.
- Helps the inference mechanism focus on important directions.

4. Acquisitional Efficiency

- Ability to acquire new knowledge automatically.
- Intelligent systems should learn new information and update their knowledge base.

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Knowledge Base

- A knowledge base stores knowledge in an organized way.
- It contains:
  - ✓Facts
  - ✓Rules
- Facts represent specific truths about the world.
  Example: “VIT is a deemed university.”
- Rules describe how to derive new facts from existing ones.
  Example: “All triangles have three sides.”

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Knowledge Base

Types of Knowledge Bases

1. Machine Readable Knowledge Base

- Stores knowledge in computer readable format.
- Uses automated deductive reasoning.
- Often represented in rule form.

2. Human Readable Knowledge Base

- Designed for humans to search and use.
- Includes articles, manuals, FAQs, and search engines.
- Example: A web search engine where users browse information.

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Issues in Knowledge Representation

The main goal of knowledge representation is to help in drawing conclusions.

- Some issues are:

Important Attributes

- Two important attributes:
- Instance
- IS-A

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Issues in Knowledge Representation

Relationship Among Attributes

- Four important properties:
- Inverse – checks consistency when a value is added.
- IS-A hierarchy – generalization and specialization.
- Reasoning about values – height, weight, age etc.
- Single valued attribute – unique value.
- Example: President of India.

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Issues in Knowledge Representation

Choosing Granularity

- Decide the level of detail for representation.
- Primitives are basic concepts like seeing, playing, holding.
- It is difficult to decide which primitives to use.
- Example:
  "John spotted Jolly."
  Spotted(agent(john), object(jolly))

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Issues in Knowledge Representation

Representing Sets of Objects

- Some properties are true for a set but not for individual members.
- Two methods:
- Extensional definition – list all members of the set.
- Rule-based definition – define a rule to check membership.
- Example:
  "English speakers are all over the world."

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Issues in Knowledge Representation

Finding the Right Structure

- Selecting the correct structure is important.
- Questions to consider:
- How to select the most suitable structure?
- How to add details from the current situation?
- What if the chosen structure is not suitable?
- When should we create a new structure?
- There is no perfect general solution.

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Issues in Knowledge Representation

Example:
“John went to Ambroiza. He ordered pizza. He paid and left.”
Question: Did John eat dinner?
Answer: Yes (inferred logically).

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THANK YOU

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